

TIM HORTONS

Business Performance Dashboard

Documentation Project Report

A professional technical and business documentation package covering the dataset, dimensional model, KPI framework, dashboard architecture, analytical findings, recommendations and future enhancements.

17,444	Orders	43,529	Units Sold
\$254.58K	Revenue	\$95.61K	Profit
37.6%	Profit Margin	\$14.59	Average Order Value
6	Stores	5	Cities

Project purpose: Transform transactional sales data into an interactive Power BI solution for monitoring revenue, profitability, customer behavior, product/category performance and store/geographic performance.

Tooling: Microsoft Power BI • Power Query • DAX • Excel • Star-schema data modeling

1. Executive Summary

This project converts a 2024 transactional sales dataset into a multi-page Power BI business intelligence dashboard. The solution uses a dimensional model so users can analyze performance by date, store, location, product/category and customer type.

The report contains four analytical views: **Executive Overview**, **Product & Category Analysis**, **Customer & Sales Behavior**, and **Store & Geographic Analysis**. The navigation moves from executive monitoring to operational diagnosis and detailed business analysis.

Headline result: 17,444 orders generate \$254,579.62 in revenue and \$95,611.88 in profit, producing a 37.6% margin. Coffee contributes \$131.62K (51.7%) of revenue; returning customer-type transactions contribute \$164.51K (64.6%); Toronto contributes \$135.13K (53.1%).

Business Questions Addressed

- How much revenue, profit, order volume and unit volume are being generated?
- Which categories and products drive the greatest revenue and profit?
- How does performance vary by month, weekday, time of day and payment method?
- What is the relative contribution of new versus returning customer transactions?
- Which stores, cities and regions are responsible for the strongest performance?
- Where are the clearest opportunities for retention, cross-selling and geographic growth?

2. Data Source & Dataset Profile

The supplied workbook contains one transactional worksheet, **SalesData**, with 17,444 records and 22 fields. The Power BI solution enriches the fact data with dimensional tables for date, store, product and location.

Attribute	Value
Source file	Tim_Hortons_Portfolio_Dataset(1).xlsx
Fact table	SalesData
Rows / transactions	17,444
Date coverage	01-Jan-2024 to 30-Dec-2024
Stores / cities	6 stores / 5 cities
Categories / products	5 categories / 19 products
Customer types	New, Returning
Payment methods	Credit Card, Debit Card, Cash, Mobile Wallet
Core measures	Revenue, Cost, Profit, Quantity, Discount, UnitPrice

Important interpretation: the source has OrderID but no dedicated CustomerID. Consequently, the dashboard's "customer" KPIs are transaction/order-based counts by customer type, not unique individual customers.

3. Data Dictionary

Field	Meaning	Type
OrderID	Unique transaction/order identifier	Integer
OrderDate	Transaction date	Date
OrderTime	Transaction time	Time
Year / Quarter / Month	Calendar attributes for time analysis	Date/Text

Day / DayName / Hour	Day and time attributes	Integer/Text
Store / City	Store and city descriptors	Text
Category / Product / Size	Product hierarchy descriptors	Text
Quantity	Units sold in the transaction	Integer
UnitPrice	Price per unit	Decimal
Discount	Discount amount/value	Decimal
Revenue	Sales revenue	Decimal
Cost	Transaction cost	Decimal
Profit	Revenue less cost	Decimal
PaymentMethod	Payment channel	Text
CustomerType	New or Returning classification	Text

4. Data Model & BI Architecture

The model follows a star-schema pattern. **SalesData** is the central transactional fact table, while descriptive dimensions provide controlled filtering and grouping. This design is cleaner and more scalable than relying on one flat reporting table.

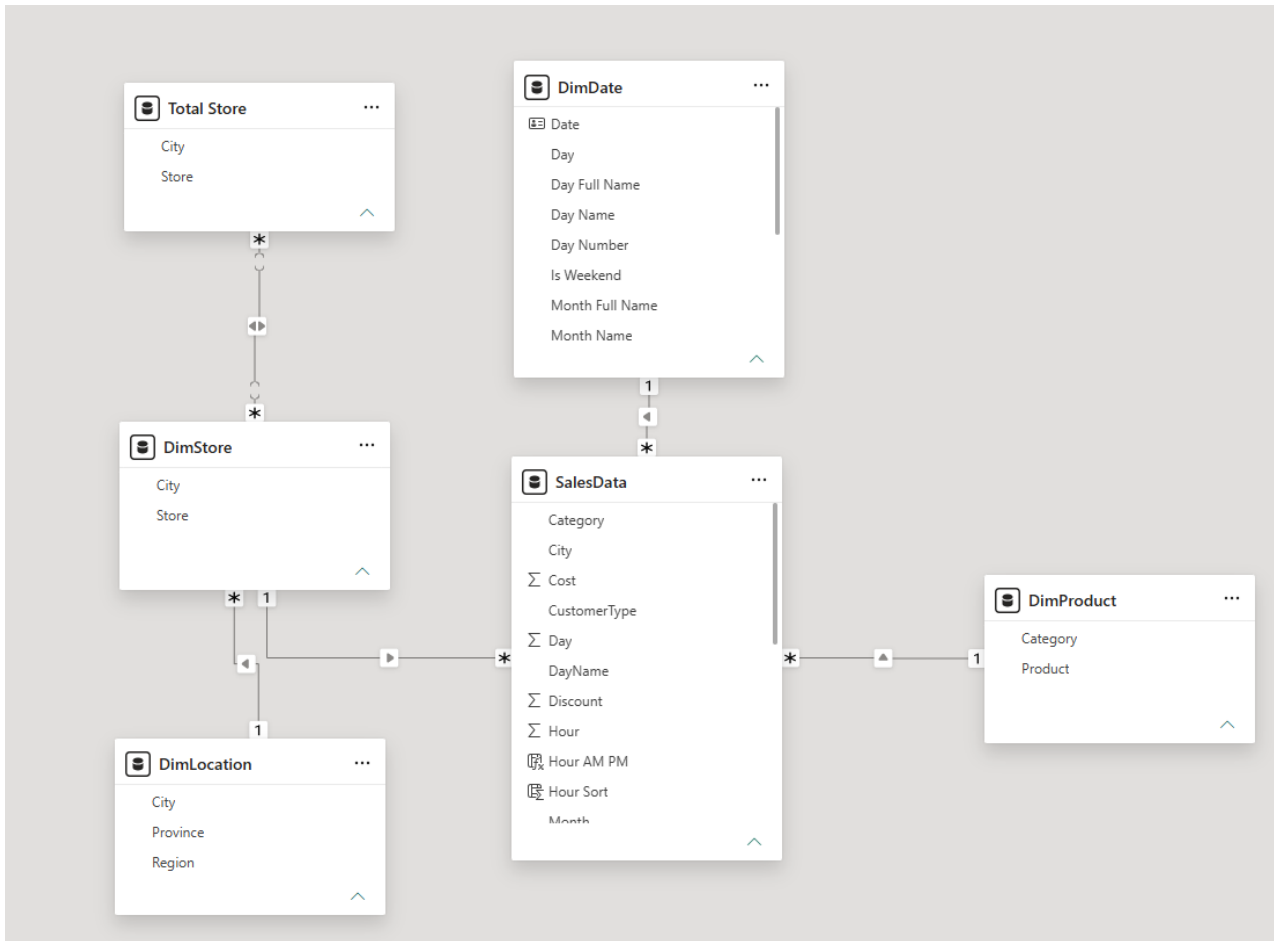


Figure 1. Power BI relationship/model view supplied with the project.

Model object	Purpose	Relationship / role
SalesData	Central fact table containing transaction-level sales, cost, profit, quantity and date/time data.	Many-side fact
DimDate	Date, day, weekday, month and weekend attributes.	1 : * → SalesData
DimProduct	Product/category lookup.	1 : * → SalesData
DimStore	Store and city lookup.	1 : * → SalesData
DimLocation	City, province and region attributes.	Geographic enrichment
Total Store	Supporting lookup/helper visible in the model.	Helper structure

Modeling principles: dimensions on the one-side, SalesData on the many-side; reusable measures for KPIs; dedicated calendar attributes; and controlled filtering through dimensional relationships.

5. KPI & Measure Definitions

KPI	Definition / logic	Business purpose
Total Revenue	SUM(SalesData[Revenue])	Top-line performance
Total Profit	SUM(SalesData[Profit])	Absolute profitability
Total Orders	DISTINCTCOUNT(SalesData[OrderID])	Transaction volume
Units Sold	SUM(SalesData[Quantity])	Product volume

Profit Margin	$\text{DIVIDE}([\text{Total Profit}], [\text{Total Revenue}])$	Profitability efficiency
Average Order Value	$\text{DIVIDE}([\text{Total Revenue}], [\text{Total Orders}])$	Spend per transaction
Average Price	$\text{DIVIDE}([\text{Total Revenue}], [\text{Units Sold}])$	Average realized price
New / Returning	Order count by CustomerType	Customer mix proxy

Validated KPI values: Revenue \$254,579.62 • Profit \$95,611.88 • Orders 17,444 • Units 43,529 • AOV \$14.59 • Average price \$5.85 • Profit margin 37.6%.

6. Data Preparation & Quality Checks

- Validated the workbook structure: one SalesData worksheet with 17,444 rows and 22 columns.
- Checked numeric fields used for aggregation: Revenue, Cost, Profit, Quantity, UnitPrice and Discount.
- Confirmed date/time attributes support monthly, weekday and time-of-day reporting.
- Separated descriptive entities into dimensions for model-driven filtering.
- Used chronological sorting logic for calendar and hour-based visuals where required.
- Cross-checked headline totals against the supplied dashboard screenshots.

Quality limitation: without CustomerID, true customer-level retention, churn, purchase frequency and lifetime value are not measurable.

7. Dashboard Design & Navigation

The report uses a consistent Tim Hortons-inspired visual system: cream canvas, red accents, rounded KPI cards, compact slicers, icon-led navigation and a persistent footer. The design prioritizes executive readability while retaining detailed analysis on dedicated pages.

7.1 Executive Overview



Figure 2. Executive Overview — headline financial and operating performance.

- **KPI layer:** revenue, profit, orders, units sold and margin.
- **Trend layer:** monthly revenue/profit movement across 2024.
- **Performance layer:** store and category rankings.
- **Product layer:** top-five products by revenue.

7.2 Product & Category Analysis

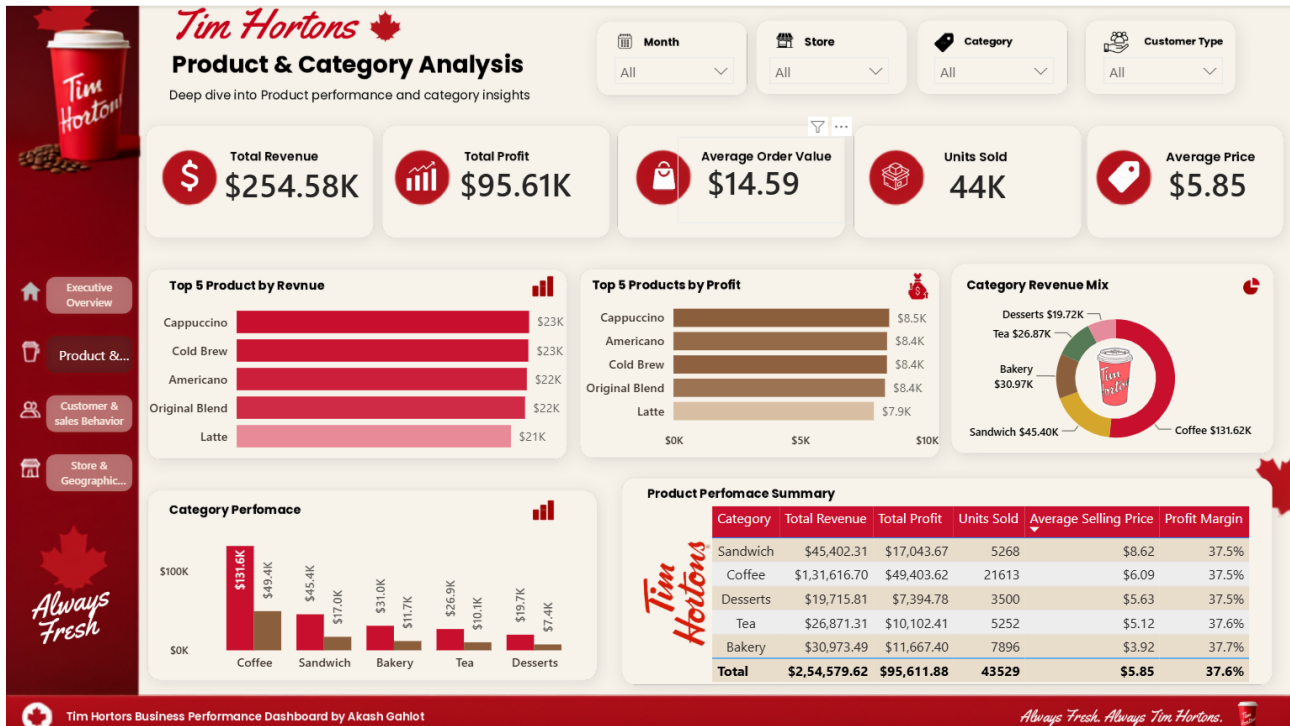


Figure 3. Product & Category Analysis — mix, rankings and category summary.

- Top-five revenue and profit products support assortment and promotion decisions.
- Category Revenue Mix highlights portfolio concentration.
- Category Performance compares revenue, profit, units, average selling price and margin.

7.3 Customer & Sales Behavior



Figure 4. Customer & Sales Behavior — transaction mix, timing and retention proxy.

- Returning transactions generate substantially more revenue than new-customer transactions.
- Weekday and time-of-day visuals reveal when demand is concentrated.
- Payment-method profitability highlights the role of card/digital transactions.
- Monthly returning-vs-new revenue percentages provide a directional retention proxy.

7.4 Store & Geographic Analysis

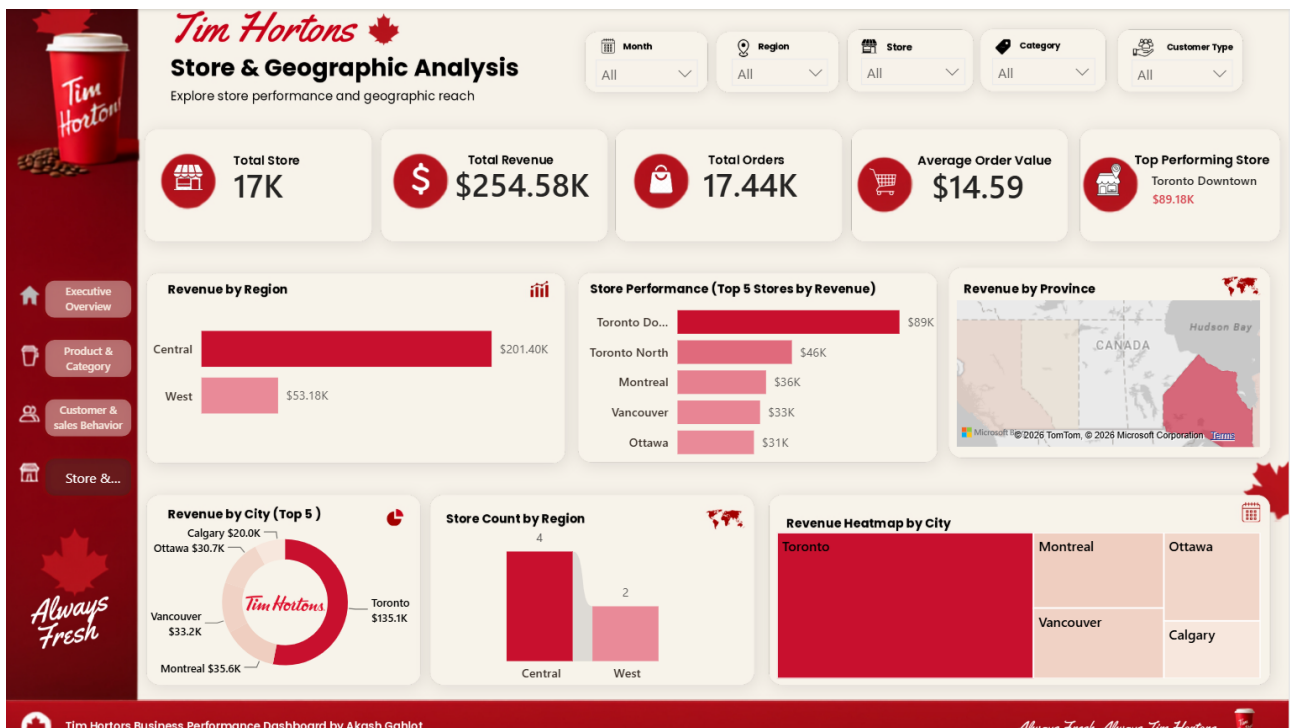


Figure 5. Store & Geographic Analysis — regional, city and store performance.

- Regional and city views identify geographic concentration.
- Top-store ranking identifies locations to benchmark.

- Map and heatmap visuals provide geographic context.

8. Key Business Insights

Finding	Evidence	Business implication
Coffee is the core revenue engine	Coffee generates \$131.62K, or 51.7% of revenue.	Protect availability and use coffee-led bundles to drive attachment sales.
Returning transactions dominate	Returning transactions generate \$164.51K, or 64.6% of revenue.	Retention and repeat-visit programs can materially support revenue stability.
Toronto is the largest market	Toronto generates \$135.13K, or 53.1% of revenue.	Benchmark Toronto practices and assess whether they can transfer to other markets.
Toronto Downtown leads	Toronto Downtown generates \$89.18K, about 35.0% of total revenue.	Investigate traffic, assortment, timing and operating practices behind the outperformance.
Saturday is the strongest day	Saturday generates \$38.06K in revenue.	Align staffing and inventory with weekend demand.
Credit card is the leading payment channel	Credit-card revenue is \$137.54K, or 54.0%.	Card/digital-first offers and loyalty integration can be prioritized.

9. Business Recommendations

- **Strengthen retention:** build repeat-visit offers around the returning customer base.
- **Optimize category mix:** protect coffee availability while using sandwiches, bakery and desserts as attachment opportunities.
- **Replicate top-store practices:** benchmark Toronto Downtown against other locations on product mix, timing and transaction volume.
- **Plan labor and inventory by demand:** use weekday/time-of-day patterns for staffing and replenishment.
- **Use geographic segmentation:** treat Toronto/Central as a benchmark while evaluating West-region growth.
- **Improve customer analytics:** add persistent CustomerID so true retention, frequency, churn and lifetime value can be calculated.

10. Technical Implementation Notes

Power Query: import the workbook, validate data types, standardize date/time fields and prepare dimension attributes.

Data model: use SalesData as the central fact table and connect DimDate, DimProduct, DimStore and DimLocation through stable keys/attributes.

DAX: use reusable measures for Revenue, Profit, Orders, Units Sold, Profit Margin, AOV and Average Price. Use DIVIDE for ratios and DISTINCTCOUNT for order counts.

Visualization: cards for KPIs, lines for trends, ranked bars for comparisons, donuts for mix, scatter for revenue/profit positioning, and map/heatmap visuals for geography.

UX: synchronized slicers, page navigation, descriptive titles and consistent formatting make the report easy to explore.

11. Portfolio / Delivery Structure

Asset	Purpose
Tim Hortons.pbix	Interactive Power BI report containing model, measures, visuals and navigation.
Tim_Hortons_Portfolio_Dataset(1).xlsx	Source transactional dataset.
Documentation Project Report.pdf	Business and technical documentation for the portfolio project.
Dashboard screenshots	Static evidence of the four report pages and model.

12. Limitations & Future Enhancements

- Add CustomerID for true unique-customer analytics and retention measurement.

- Add targets/budgets for variance and attainment analysis.
- Add inventory, staffing and promotion data for operational root-cause analysis.
- Add standardized geographic keys/coordinates for stronger map accuracy.
- Add additional years for year-over-year and seasonality analysis.
- Introduce a dedicated measures table and formal KPI dictionary for enterprise governance.

13. Conclusion

The Tim Hortons dashboard demonstrates a complete business analytics workflow: transactional data → dimensional model → reusable KPIs → interactive dashboards → business insights → recommendations. It combines financial analysis, customer behavior, product analytics and geographic performance in a single coherent reporting experience.

Overall assessment: the solution provides a management-friendly view of where revenue and profit are generated, when demand occurs, which products/categories matter most, and which locations lead performance. The next maturity step is to enrich the data with customer, target, operational and historical attributes.

Branding note: This document describes the supplied portfolio dataset and dashboard for analytical/educational purposes. It should not be interpreted as an official Tim Hortons corporate performance report unless the underlying data is independently verified and authorized.